



CONTACT

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PROFESSIONAL SUMMARY

Accomplished Lead Software Engineer with expertise in microservices architecture, cloud infrastructure, and API development. Proven track record of leading engineering teams to deliver scalable solutions and enhance platform reliability. Seeking to leverage skills in stakeholder collaboration and technical leadership to drive innovation in a dynamic software development environment.

SKILLS

- Team Leadership
- Stakeholder Collaboration
- Technical Documentation
- Microservices Architecture
- API Development
- ETL Processes
- Real-time Analytics
- Machine Learning
- Generative AI
- System Administration
- Automated Testing
- Observability
- Backend Development
- Frontend Development
- Cloud Infrastructure
- Database Administration
- Cross-Service Communication
- Fault-Tolerant Architecture
- Message Queue Systems
- Token Management
- Authentication
- Prototyping
- Scalability
- Concurrency Handling
- Distributed Caching
- Horizontal Scaling
- Data Management

MICHAEL LOWENSTEIN

LEAD SOFTWARE ENGINEER

EXPERIENCE

Senior Consultant — Fullstack Principal Engineer

Norima (Client: Millennial Specialty Insurance MGA)

01/2026 - Present

- Lead production data-integrity remediation across a multi-generational insurance policy administration platform (MSI-PAS), authoring governed SQL remediation packages submitted through a formal DBA review process covering diagnostics, dry-run validation, fix execution, and rollback scripts
- Investigate and resolve complex data corruption clusters spanning payment misapplication, snapshot flag corruption, billing schedule gaps, missing reference data, and endorsement flow failures — each categorized into a growing root-cause taxonomy (Clusters A through W)
- Author and maintain msi-nav, a data-driven CLI navigation and compliance scanning tool enabling developers to locate any operation, product, status code, or queue across five repositories in under 60 seconds, reducing onboarding ramp time and eliminating ad-hoc tribal knowledge requests
- Conduct architectural spike investigations into snapshot corruption patterns, mortgage company write failures, enum-to-database migrations, and frontend modernization via Bootstrap 5 and Vue island proof-of-concept, each documented with deep-dive rationale, generalized pattern extraction, and benchmarked current state
- Collaborate with production support, underwriting, DBA, and engineering leadership to triage, scope, and resolve zero-downtime data issues

Lead Backend Software Engineer Rogers Communications Inc. - Calgary, AB

11/2025 - Present

- Lead backend engineering for the next-generation Rogers eCommerce platform, delivering highly scalable, cloud-native services using C# and ASP.NET Core within a microservices architecture
- Manage and mentor a small senior engineering team, providing technical leadership, architectural direction, code reviews, and delivery oversight across multiple concurrent workstreams
- Own design and implementation of mission-critical commerce services including product catalog, pricing, checkout, order orchestration, and integrations with enterprise systems
- Drive platform reliability, performance, and scalability for high-traffic consumer workloads through distributed caching, asynchronous messaging, and resiliency patterns
- Collaborate cross-functionally with product, UX, platform, DevOps, and business stakeholders to translate complex requirements into robust technical solutions
- Establish engineering standards, CI/CD practices, and observability strategies to support rapid, safe deployment cycles
- Lead technical decision-making on data architecture, API contracts, security, and integration strategies across the commerce ecosystem
- Contribute hands-on to development of core services while coordinating delivery across teams in an enterprise Agile environment

Independent Contractor - Senior Software Engineer 1280 Labs Inc.

03/2025 - Present

- Team Lead on the front-end application for new AI-enabled Talent Management CRM
- Engineered a suite of backend services using Python, Node.js, and Redis
- Developed and maintained Flutter-based mobile client and Angular web dashboard
- Authored a new client-side token management and authentication route-mapping system using an Auth0-based JWT token schema
- Maintained automated testing suites and provided production-level observability
- Prototyping of Back-End services in different frameworks - Golang, Rust, Node's, Python, .NET CORE
- Architected high-throughput microservices handling concurrent user sessions and real-time data processing, implementing distributed caching strategies and horizontal scaling patterns
- Built resilient message queue systems for cross-service communication, designing fault-tolerant architectures with retry mechanisms and dead letter queues

- Angular
- React
- Vue
- Python
- Node.js
- C#
- .NET Core
- TypeScript
- JavaScript
- HTML
- CSS
- SCSS
- RabbitMQ
- REST API Design
- GraphQL
- Optimization
- Amazon SQS
- Kafka
- Rust
- PostgreSQL
- AWS
- Azure
- Express
- API development

- Led the integration of AI-driven features for device diagnostics and behavior analytics

Senior Back-End Software Engineer College House at House Analytics

09/2024 - 03/2025

- Design and implement enterprise-scale .NET Core solutions, including CQRS Web APIs, database administration, and AWS infrastructure integrations
- Led the planning, prototyping, and deployment of back-end infrastructure for analytics products, modernizing legacy systems and creating a test automation framework to enhanced efficiency and scalability
- Accomplished the design and prototyping of a state-of-the-art Message Bus architecture, exploring a range of technologies like AWS MQ/SQS, RabbitMQ, Redis, and Kafka to maximize communication effectiveness and reliability.
- Successfully integrated advanced machine learning features into production systems, leading to improved planning, presentation, testing, and deployment processes.
- Collaborated directly with stakeholders to future-proof products, ensuring alignment with business goals, and author detailed documentation for end-users and internal teams

Senior Software Engineer NCAL Engineering

01/2024 - 02/2025

- Implemented ETL processes to synchronize time-tracking APIs with production databases
- Broad stack including NestJS, NodeJS JWT Token Authentication, Python scripting for setup & teardown of short-lived servers, and Redis Cache APIs - Golang was used for heavier workflows due to performance bottlenecks
- Built and extended a custom data management warehouse for engineering graphics and metrics using Angular 16/17, .NET Core Web APIs, and supporting NodeJS microservices
- Highlights included building a Role-Based Access Control (RBAC) authentication infrastructure using Angular and .NET Core with Auth0
- Designed & deployed an Invoice Automation and Analytics Dashboard - leveraging Angular 16, React UI Caching, file storage, and database access were supported by microservice integration with central API Gateway
- Managed DevOps pipelines and deployments using AWS, in-house remote VMs, and Docker Compose
- Saved the Senior Engineer and Founder a full week of hours every month, streamlining financial workflows

Senior Software Engineer FullStack Labs

12/2021 - 12/2023

- Led the planning, design, and implementation of front-end systems for a Custom Cabinetry CRM/Project Management Application
- Identified and co-led the refactoring and integration of Functional Components into the flagship FSL Admin Application.
- Authored back-end ETL applications using Typescript, Node.js Express, and NestJS frameworks - global invoicing accuracy improved from 95.1% to 99.8%.
- Machine Learning R&D for ETL pipelines - migration of high-throughput + low-latency workflow to Golang and Rust for performance comparison.
- Engineered high-availability ETL pipelines processing financial data with 99.8% accuracy, implementing robust error handling, transaction rollbacks, and data integrity checks critical for financial services compliance
- Developed and maintained the front-end architecture for an Enterprise Association Management Portal - this was highlighted by co-authoring a large-scale migration of over 10,000 users from legacy IdentityServer to Auth0 for a flexible User, Group & Custom Authentication protocol.

Software Engineer Peloton Computer Enterprises

12/2016 - 12/2021

- Developed a new web application, ProdView, for task scheduling and operations in the oil and gas sector
- Introduced and led the frontend implementation of OAuth2 and OpenID Connect (OIDC) authentication across the company's infrastructure using Angular 7-9
- Designed and implemented Angular front-end MVPs to showcase features before full-scale planning
- Maintained and enhanced the Land Management System for oil and gas clients
- Managed Linux servers, optimized PostgreSQL and FreeBSD systems
- Rebuilt legacy systems as polished microservices/pilots using modernized tools and frameworks - NestJS, Python, Golang, Rust

- Developed distributed task scheduling systems for oil & gas operations, creating reliable job queues and worker processes that mirror the architectural patterns used in blockchain networks and cryptocurrency mining coordination
- Implemented secure API gateways and authentication middleware for financial data access, building authorization layers and rate limiting systems essential for protecting high-value financial transactions and user account security

Software Developer SICharts (SIA Wealth Management)

12/2015 - 12/2016

- Rebuilt a legacy software application using modern web technologies
- Developed a suite of financial market tools
- Utilized MSSQL Server, ASP.NET (C#), JavaScript, jQuery, Bootstrap, and Python test suites

EDUCATION

Bachelor of Science: Computer Science - Software Engineering

University of Calgary , 12/2016

Bachelor of Science: Kinesiology

University of Calgary , 05/2013

REFERENCES

References available upon request

WEBSITE, PORTFOLIO AND PROFILES

<https://linkedin.com/in/mlo2gs>

<https://www.lowenstein.ca>